

Vrund Patel

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Education

BE : Vishwakarma Government Engineering College 2022 – 2026
Computer Science & Engineering (Data Science)

- CGPA : 9.21

HSC : Swastik Education Campus(GSEB Board) 2022

- Percentage : 88.67% , Percentile : 96.83

Experience

AI Engineer Intern – iQudTek 📅 May 2026 – Present

- Designed and developed an end-to-end cloud-native Computer Vision pipeline for smoke study analysis, powering the company's product and client solutions.
- Architected and deployed the complete workflow on AWS using Lambda, SageMaker, S3, CloudWatch, and other managed services.
- Independently owned the complete development lifecycle, from model training and pipeline design to cloud deployment and production integration.

AI/ML Engineer Intern – Bacancy Technology 📅 Jan 2026 – May 2026

- Worked on end-to-end AI application development spanning ML, DL, NLP, GenAI, and Agentic AI.
- Designed workflow automations with n8n to streamline business processes and API integrations.
- Built scalable backend services using FastAPI and integrated them with React applications.

Projects

RoastForge – MCP-Enabled Agentic AI Resume Roaster & Rebuilder [🔗 RoastForge](#) [in Video](#)

- Built a **8-node** LangGraph + FastMCP agentic workflow that parses resumes, brutally roasts weaknesses, evaluates ATS fit (0–100), iteratively rebuilds until score > 90, generates interview questions, and exports professional PDF resumes autonomously
- Integrated Claude Desktop **MCP with parallel node execution** (Roast + ATS), conditional routing loops, MCP Resources for real **PDF delivery**, and structured Pydantic outputs; Increased average ATS score from 55 to 84 (+53%) through **iterative multi-agent refinement**.

Agentic RAG for Code-Mixed WhatsApp Chat Memory [🔗 WhatsUp](#)

- **Hybrid retrieval** - dense (bge-m3) + BM25, RRF fusion, bge-reranker-v2-m3 cross-encoder reranking; per-chunk LLM glosses to make romanized Gujarati embeddable.
- **LangGraph pipeline** - LLM emits a structured filter (intent / date range / sender), deterministic routing to semantic | temporal | event nodes.
- **Voice + streaming** - Whisper large-v3-turbo (Groq) speech-to-text with a transcript-confirmation gate, offline faster-whisper fallback, SSE token streaming with live pipeline trace, Ollama & Groq swappable per task.

Laminar Smoke Study Analysis (AWS)

- Build end-to-end CV pipeline for pharmaceutical smoke-study videos: scene gate, YOLO-seg smoke detection, pipe/camera gates, optical flow, and statistical laminar baseline scoring.
- Designed 10-metric deviation engine with explainable verdicts, visual evidence(ROI, Voids, Annotated frames/clips), and human-feedback suppression.
- **Deployed** GPU inference **on SageMaker** with **S3 job queue** and **Lambda triggers** for upload-to-results workflow.

Technical Skills

Languages: Python, SQL

Cloud & MLOps: AWS (SageMaker, Lambda, S3, CloudWatch, IAM, EC2)

GenAI & Agentic AI: LangChain, LangGraph, LangSmith, RAG, MCP, FAISS, Prompt Engineering, Hugging Face

Computer Vision: OpenCV, YOLO, Roboflow, Transfer Learning, Fine Tuning, Data Annotation, Object Detection, Image Classification, Image Segmentation

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, CNNs, LSTMs, Transformers, Feature Engineering, Model Evaluation, NLP

Data Science: Pandas, NumPy, Matplotlib, Power BI

Developer Tools: Git, GitHub, Jupyter, Google Colab, n8n

Familiar With: FastAPI, React, R, Java, JavaScript